

NEXPIRE

NEAR EXPIRY INTELLIGENCE

TEAM NAME : SPARTANS

TRACK : SOCIAL IMPACT & ACCESSIBILITY

COLLEGE: MAHARAJA SURAJMAL INSTITUTE OF TECHNOLOGY

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THE EXISTING SYSTEM – PROBLEM IN INDIA



India discards an estimated **78.1 million tonnes** of food every year — the **highest absolute figure** of any country after China, while roughly **17%** of all food produced globally is wasted.

Retail-level waste in India alone is estimated at 5–16 million tons annually. – (NEXPIRE Playbook, Ch.1)



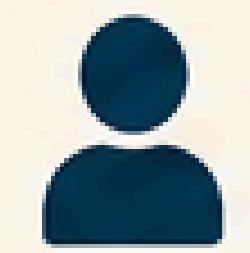
WHY IT KEEPS HAPPENING (ROOT CAUSES)



Static, manual markdowns — stores follow fixed rules (“20% off at 4 days, 40% at 2 days”) regardless of real demand, weather, or footfall



Blind to real-time signals — a heatwave killing dairy sales or a rainy evening cutting footfall isn’t factored into pricing



Manual, staff-dependent decisions — markdown calls made department-by-department, inconsistent across shifts/stores



No real-time donation channel — surplus-to-shelter coordination still happens over phone calls, missed pickup windows



No proactive alert to nearby buyers — bargain shoppers/NGOs have no way to know surplus exists nearby, right now



WHO SUFFERS?



STORES



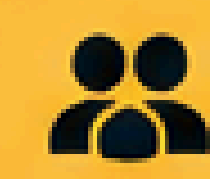
Shrinkage eats thin perishable margins



Staff hours lost to manual markdown calls



No structured data for ESG/CSR reporting



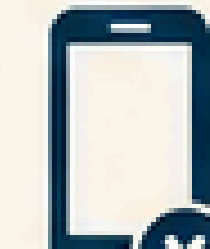
CONSUMERS & COMMUNITY



No visibility into real nearby discounts



Shelters get unpredictable, last-minute supply

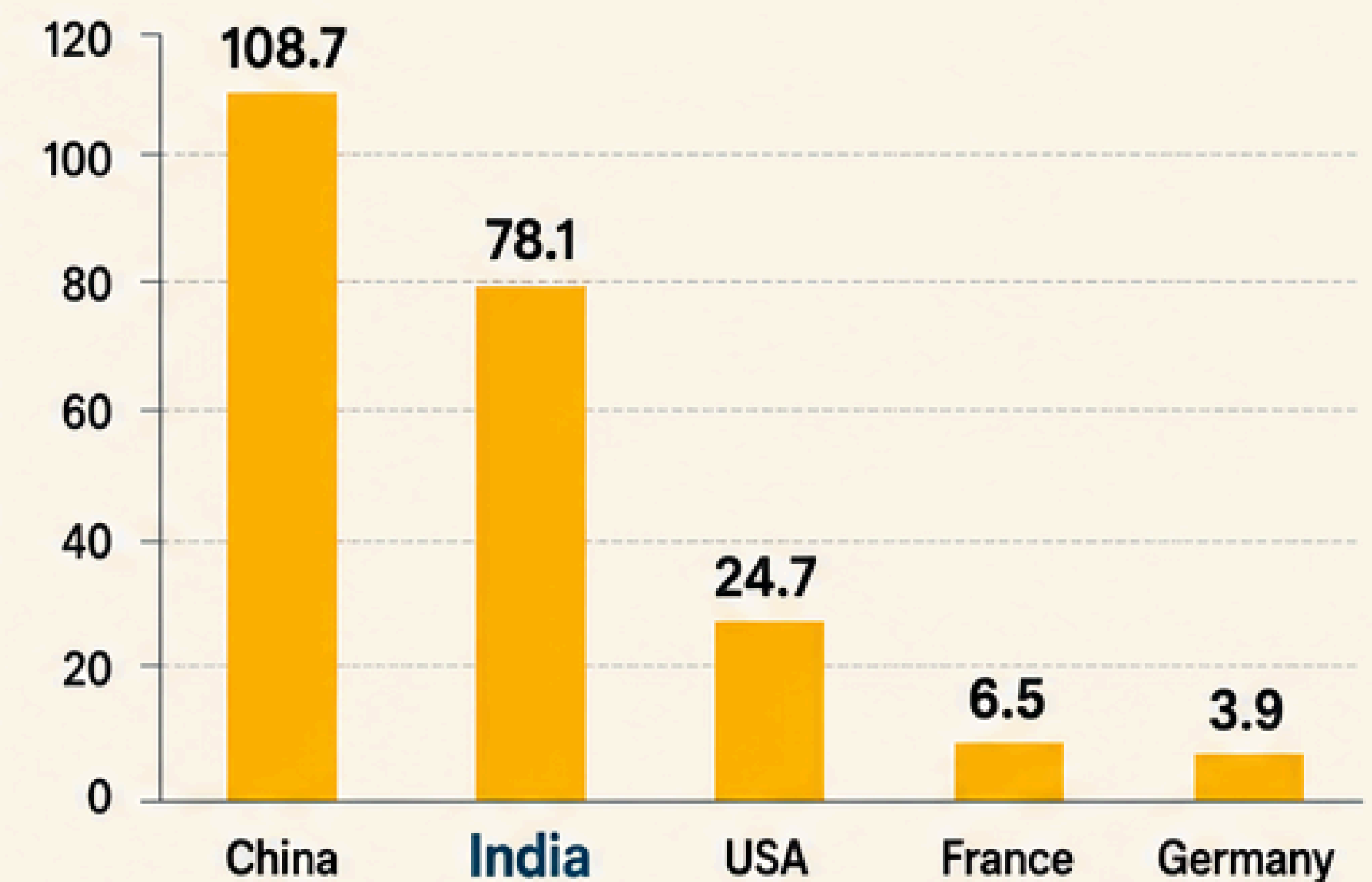


Non-smartphone users excluded from app-first solutions



FOOD WASTE BY COUNTRY

(MILLION TONNES / YEAR)



China wastes an estimated **108.7** million tonnes of food per year while India discards **78.1** million tonnes, and the United States creates **24.7** million tonnes annually.





THE SPECIFIC BREAKPOINT WE IDENTIFIED



The system doesn't fail at "too much food" — it fails at one narrow, specific moment: the **4–8 hour window** between an item being flagged as at-risk and it actually being sold, donated, or thrown out.

Nothing connects those two ends in real time.



BREAK 1 — DETECTION → PRICING



Markdown is decided on a **fixed calendar** (e.g. "40% off at 2 days out"), not on what's actually happening — a slow Tuesday and a busy Saturday get the **same treatment**.

BREAK 2 — PRICING → AWARENESS



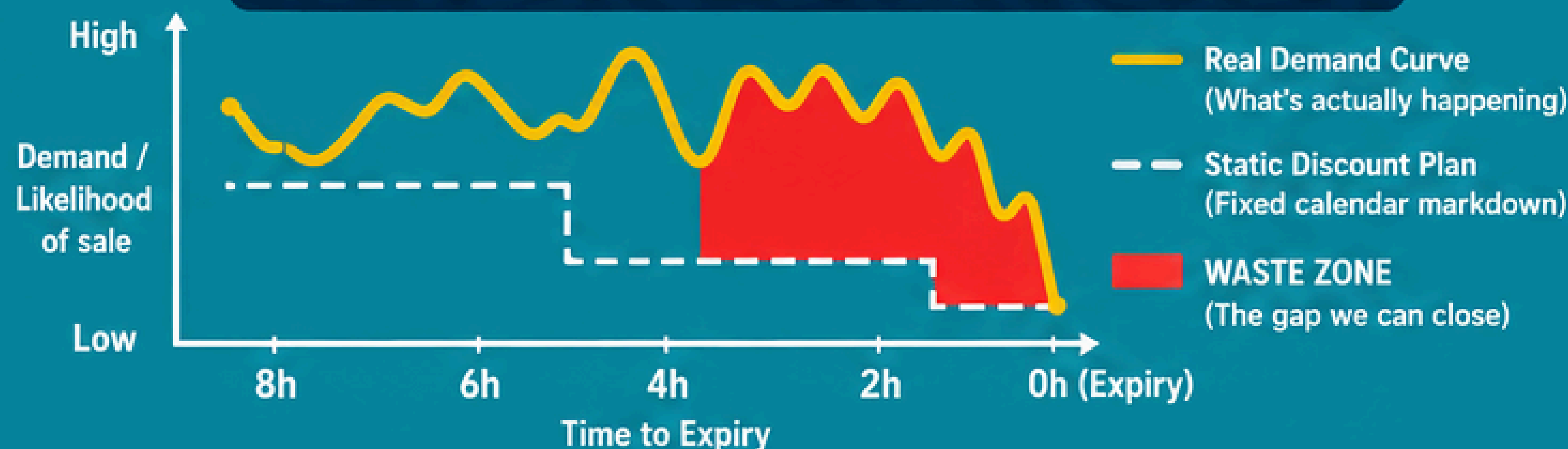
Even when a real discount exists, there's **no channel** broadcasting it to nearby buyers or shelters at the moment it's created — it **sits on a shelf, undiscovered**.

BREAK 3 → AWARENESS → REDISTRIBUTION



When stores do want to donate, coordination happens over phone calls with **no fixed pickup window** — recoverable food still gets thrown out due to timing, not lack of intent.

ITEM LIFECYCLE: EXPIRY COUNTDOWN VS. ACTION TAKEN



WHY THIS SPECIFIC BREAKPOINT MATTERS



It's a **timing failure**, not a supply or demand failure — the food, the buyers, and the shelters already exist.



It's the one point in the chain that's **fully automatable** with data already sitting in a store's inventory system.



Closing this single gap is what **every competitor** (Too Good To Go, Flashfood, Wasteless) leaves at least partially open — each owns one side of the break, none connect all three.

THE SCALE (ESTIMATES)



60–70%

of at-risk inventory never gets a real-time alert.
(Industry Estimate)



3–6 HOURS

average donation coordination time via phone calls.
(Industry Estimate)



5–15 MINUTES

coordination & acceptance time with NEXPIRE.
(Target Outcome)

THE DIFFERENCE?
HOURS OF FOOD SAVED EVERY DAY.





WHO IS AFFECTED – AND WHY IT MATTERS



This isn't a single-stakeholder problem — **four groups lose simultaneously** from the same broken window, which is exactly why fixing it has **outsized impact**.

STAKEHOLDER	WHAT THEY LOSE TODAY	WHAT THEY EXPECT
 GROCERY & RETAIL STORES	- Shrinkage eats directly into thin perishable margins - Staff hours burned on manual markdown calls - No data trail for ESG/CSR reporting	 Automated, low-effort markdown decisions that protect margin without a costly POS overhaul
 PRICE-SENSITIVE CONSUMERS	- No visibility into real, steep discounts happening nearby, right now - Clearance sections are hit-or-miss	 Real-time, trustworthy alerts with a frictionless way to claim — ideally without installing another app
 SHELTERS & NGOS	- Unpredictable, manually-coordinated supply - Missed pickup windows - Can't plan meals around one-off scrambles	 Predictable, prioritized, free/subsidized access to surplus, with advance notice
 MUNICIPALITIES / ESG STAKEHOLDERS	No verifiable data on food diverted or emissions avoided to back sustainability targets	 Structured impact data that supports policy goals (e.g. SDG 12.3)

ONE BROKEN LOOP. FOUR LOSERS.

STORE LOSS

Shrinkage, manual effort, and no data trail eat into margins and accountability.



CONSUMER LOSS

Missed discounts and no visibility into real-time opportunities nearby.



NGO LOSS

Unpredictable supply, missed pickups, and last-minute scrambles.



CITY / ESG LOSS

No verifiable impact data to prove sustainability targets or track resource savings.



A store's loss is a shelter's missed meal, a shopper's missed discount, and a city's unmet ESG target — all from the same unsold item.

WHY IT MATTERS



These four groups aren't separate problems — they're **one broken loop**: a store's loss is a shelter's missed meal and a shopper's missed discount and a city's unmet ESG target, all from the same unsold item.



India's scale makes this urgent, not theoretical: **78.1 million tonnes** wasted annually, against a country where the FAO has ranked India among the highest for undernourishment — waste and hunger are happening in the same supply chain, not separate ones.



Every existing solution (Too Good To Go, Flashfood, Wasteless) serves one of these four groups well — **none connect all four** in a single pipeline, which is the structural gap NEXPIRE is built to close.





YOUR PROPOSED SOLUTION – NEXPIRE



NEXPIRE is a **closed-loop, hyper-local liquidation-and-redistribution engine** that sits on top of a store’s existing inventory data — it doesn’t just flag waste, it prices it, targets it, and moves it before it’s lost.



WHAT IT ACTUALLY DOES



Continuously scores every perishable batch against its **expiry timeline**.



Layers in real-world signals — weather, historical demand, foot traffic — through an **ML model** to decide two things a static system can’t: how much to discount and who to target first (consumers vs. B2B/NGO bulk buyers).



When a batch crosses a critical threshold, automatically fires a targeted **SMS/WhatsApp flash-sale alert** within a geofenced radius, takes payment, and routes fulfillment — while shelters on standing orders get subsidized, no-payment priority access.



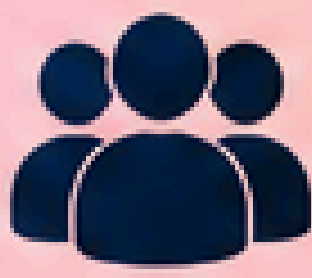
WHAT MAKES IT STRUCTURALLY DIFFERENT



Predictive, not reactive pricing
Discount computed live from a model, not a fixed markdown calendar.



No-app-required distribution
SMS/WhatsApp-first claiming — including reply-“YES” — removes India’s biggest adoption barrier: getting someone to install an app.



Dual-sided targeting
Same engine decides consumer sale vs. NGO/shelter routing — most competitors pick one lane.



End-to-end loop
Detection → pricing → alert → reservation → payment → fulfillment → impact reporting, all in one pipeline.



WHY THIS FITS INDIA SPECIFICALLY



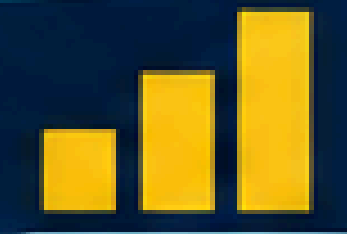
WhatsApp-first claiming maps directly to India’s messaging habits — far higher reach than a dedicated app in tier-2/3 markets.



Lightweight ingestion (CSV/photo/manual entry) means no POS overhaul — critical since most Indian retailers lack electronic shelf labels or deep inventory integration.



Directly addresses India’s **5–16 million tons** of annual retail-level waste (Ch.1) at the exact point it’s created — the store shelf, in real time.



HOW NEXPIRE STACKS UP

FEATURE	Consumer Marketplace Apps	B2B Pricing Tools	NEXPIRE
Consumer sale	✓	—	✓
B2B / NGO routing	—	✓	✓
Dynamic pricing	—	✓	✓
WhatsApp-first claiming	—	—	✓
End-to-end automation	—	—	✓



HOW IT WORKS



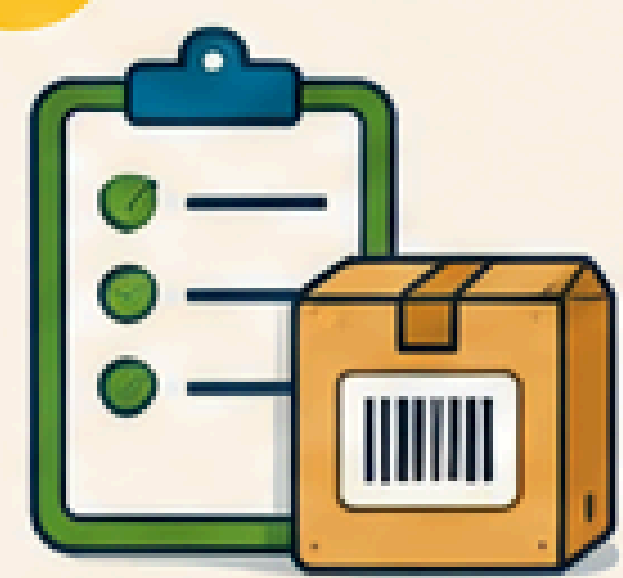
One **continuous loop** — from a store's inventory to a claimed item — running **automatically**, with **no manual markdown calls** and **no app install** required for the end user.



THE CORE LOOP, IN ONE LINE:

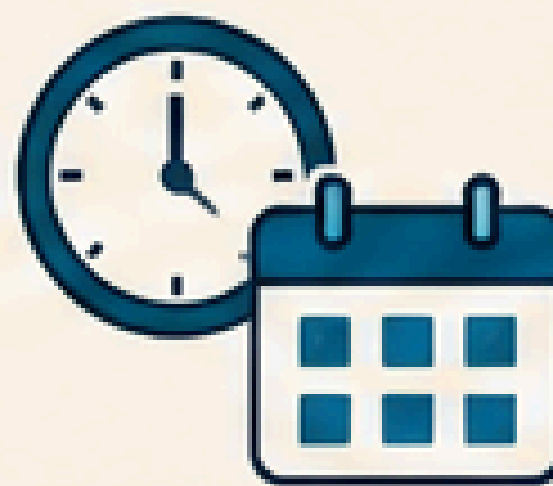
A scheduled scan finds at-risk batches → the ML model prices and segments them → an alert fires to a geofenced audience → a claim reserves stock → payment confirms it → fulfillment happens → the outcome feeds back into the model for next time.

1 INGEST



Store inventory enters via CSV upload, barcode scan, or shelf-label photo (OCR) — zero-friction onboarding, no POS overhaul needed.

2 SCAN



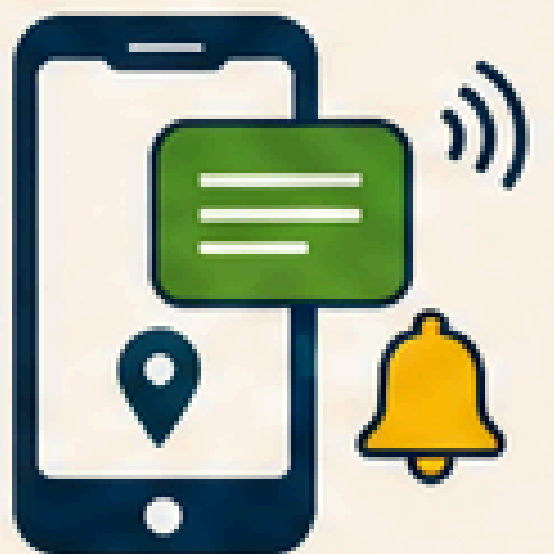
A scheduled scan (every few minutes) flags batches approaching expiry.

3 PRICE & SEGMENT



The ML model computes a live discount using weather, demand history, and foot traffic — and decides whether the batch goes to consumers or B2B/NGO bulk buyers.

4 ALERT



If a threshold is crossed, an SMS/WhatsApp broadcast fires within a geofenced radius around the store.

5 CLAIM



User taps the link or replies "YES" — no account, no app install; item is reserved with a countdown timer.

6 PAY & FULFILL



Checkout via Stripe/Razorpay test flow, then pickup (QR code) or delivery.



7 LEARN



Outcome (sold / expired / donated) logs back into the system, feeding the next model retrain and the impact dashboard.



PARALLEL TRACK — NGOS / SHELTERS



Shelters on standing orders skip the general broadcast entirely.



Notified first on their own priority channel.



Subsidized or free access.



Whenever a matching batch appears.



WHY THIS MATTERS FOR THE PITCH

Every other player automates **one** step of this loop (pricing, or alerts, or a marketplace) — NEXPIRE is the only one where **all seven steps** run as a **single, automatic pipeline**.



PRICING (Only)



ALERTS (Only)



MARKETPLACE (Only)



NEXPIRE (ALL 7 STEPS, ONE PIPELINE)



YOUR PROPOSED SOLUTION & ITS IMPACT

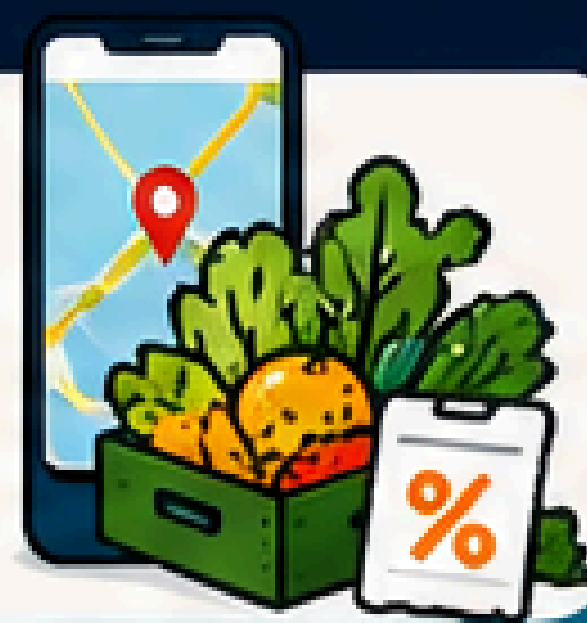
One engine. Zero waste. Real impact.



YOUR PROPOSED SOLUTION



is a closed-loop, hyper-local liquidation-and-redistribution engine that sits on top of a store's existing inventory data — it doesn't just flag waste, it **prices it**, **targets it**, and **moves it** before it's lost.



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Continuously **scores every perishable batch** against its expiry timeline.



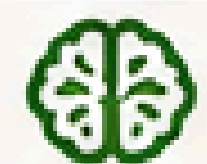
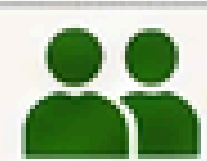
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WHAT MAKES IT STRUCTURALLY DIFFERENT

ANGLE	WHY IT CLOSSES THE GAP
 Predictive, not reactive pricing	Discount computed live from a model, not a fixed markdown calendar.
 No-app-required distribution	SMS/WhatsApp-first claiming — including reply-“YES” — removes India's biggest adoption barrier: getting someone to install an app for one discount.
 Dual-sided targeting	Same engine decides consumer sale vs. NGO/shelter routing — most competitors pick one lane.
 End-to-end loop	Detection → pricing → alert → reservation → payment → fulfillment → impact reporting, all in one pipeline.

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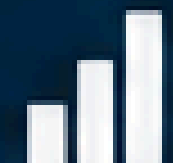


Directly addresses India's **5–16 million tons** of annual retail-level waste (Ch.1) at the exact point it's created — the store shelf, in real time.



HOW IT IMPROVES THE EXISTING SYSTEM

doesn't add a new layer on top of the old system — it **replaces the three points where the old system breaks, one for one**.

BREAKPOINT	EXISTING SYSTEM	FIXED BY OUR SOLUTION
 Static markdown calendar	✗ Fixed markdown calendar (“40% off at 2 days out”) regardless of demand.	✓ Live ML-computed discount based on weather, demand, foot traffic — different price for a slow Tuesday vs. a busy Saturday.
 Manual markdown decisions	✗ Markdown decisions made manually, department-by-department.	✓ Automated scan + pricing, freeing staff hours for customer-facing work.
 No real-time channel to nearby buyers	✗ No real-time channel to nearby buyers — clearance is hit-or-miss.	✓ Geofenced SMS/WhatsApp broadcast the moment a threshold is crossed.
 Inconsistent donation flow	✗ Donation coordination via phone calls, inconsistent pickup windows.	✓ NGO standing orders — automatic, prioritized, no-payment routing.
 No data trail for impact	✗ No structured data trail for waste/impact reporting.	✓ Built-in Eco-Impact dashboard — dollars saved, lbs rescued, CO ₂ e avoided.
 App install barrier	✗ App-install required for most existing consumer solutions.	✓ No-app-required claiming — tap a link or reply “YES” by text.



WHERE THIS SPECIFICALLY BEATS NAMED COMPETITORS

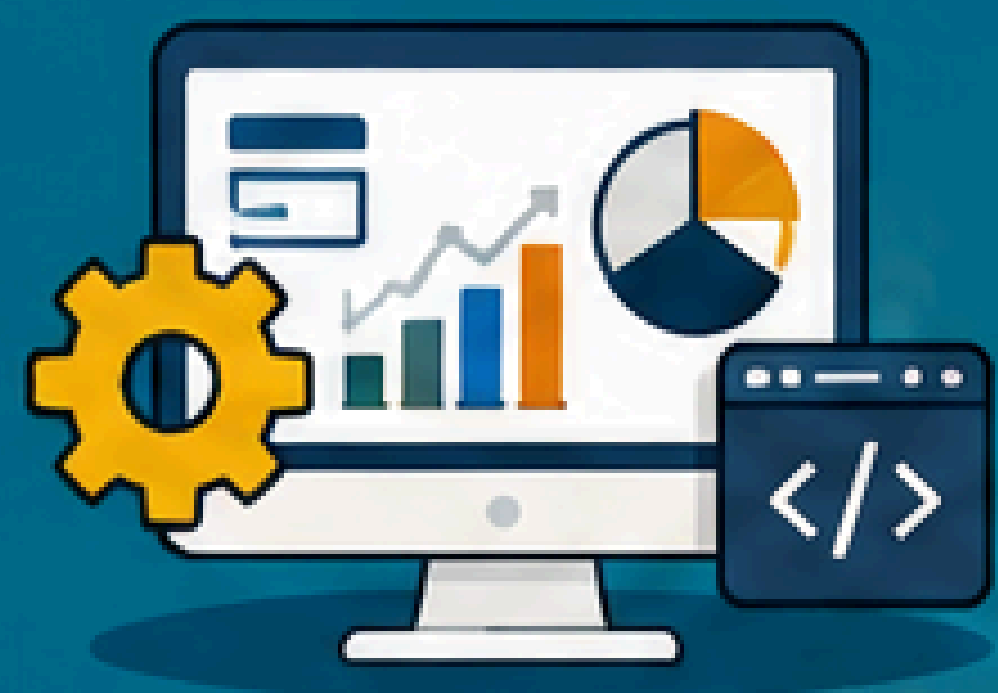
vs. Too Good To Go / Flashfood	CONSUMER MARKETPLACE APPS (e.g., Too Good To Go, Flashfood)	B2B PRICING TOOLS (e.g., Revionics, ShelfWise)	OUR SOLUTION
	 Predictive Pricing (ML)	✗	✓
	 Consumer Distribution (SMS / WhatsApp)	✓	✓
	 NGO / Shelter Routing	✗	✓
	 End-to-End Loop (Detection → Impact)	✗	✓

Their discounts are still human- or rule-s et; our markdown is **computed, not chosen**.



is the only engine that connects all three lanes — **pricing, people, and purpose** — in one automatic pipeline.





BASIC IMPLEMENTATION APPROACH



NEXPIRE is built as **three cooperating planes**, not one monolith — each buildable and demoable independently, with a lightweight ingestion path so **no retailer needs a costly POS overhaul** to pilot it.



CORE TOOLS

1 DATA & PREDICTION

Ingest inventory, enrich with weather/demand, run the ML pricing engine.

PostgreSQL + PostGIS, Scikit-Learn

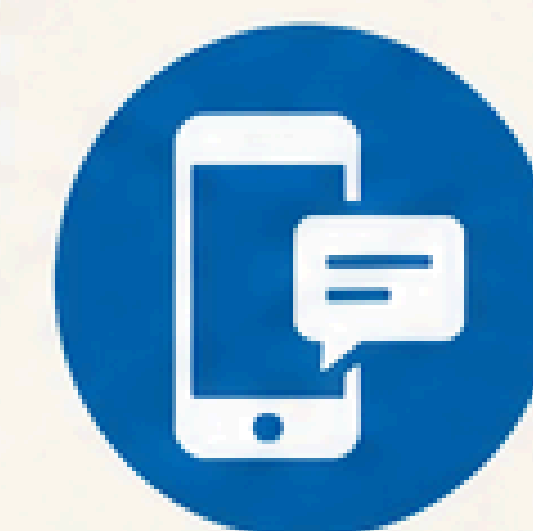


CORE TOOLS

2 ORCHESTRATION

Reservations, locking, payment state, standing-order routing.

FastAPI, Redis, Celery

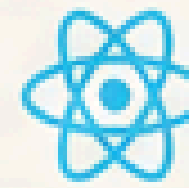


CORE TOOLS

3 DELIVERY

Everything the end user sees or receives.

Twilio (SMS/WhatsApp),
Razorpay, React



PHASE-BY-PHASE BUILD PLAN



Data & Prediction



Orchestration



Delivery

ESTIMATED EFFORT (HOURS)

0 10 20 30 40 50 60 70 80

0		SETUP	Repo, docker-compose, DB schema, seed data
1		INVENTORY CORE	CSV/manual ingestion, batch schema live
2		MARKDOWN ENGINE v1	Rule-based pricing first, then swap in the trained ML model
3		GEOFENCING & CLAIMS	PostGIS radius queries, tokenized claim link, reservation lock
4		COMMS	Twilio SMS dispatch + inbound "reply YES"; WhatsApp as stretch
5		PAYMENTS & FULFILLMENT	Razorpay test-mode checkout, pickup/delivery toggle, QR code
6		DASHBOARD & POLISH	Eco-Impact dashboard, NGO standing-order UI
7		DEMO REHEARSAL	Seeded dataset, run-through, fallback recording



Each phase is independent and value-delivering. You can stop at **Phase 3** and still have a powerful core demo.



WHY THIS ORDER?

Phases 0–3 alone prove the **core differentiator** — predictive pricing + geofenced claim — so if time runs short, that's the **fallback slice** that still demos well.



Predictive Pricing



Geofenced Alert



Easy Claim



Reserved Stock

CORE DIFFERENTIATOR PROVEN IN PHASES 0–3

INDIA-SPECIFIC IMPLEMENTATION CHOICES



Razorpay over Stripe as the default payment rail — pairs naturally with **WhatsApp Business API**, which sees far higher engagement in India than a dedicated app.



CSV/photo/manual ingestion as the entry point — most Indian retailers lack electronic shelf labels or deep POS integration, so this **removes the adoption barrier** on day one.



Synthetic, seeded dataset for the ML model where real retailer data isn't available yet — stated **openly as synthetic** for credibility, not hidden.